

E.4 Fission

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **24 marks**

$c = 3.0 \times 10^8 \text{ m s}^{-1}$; $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$. Show your working.

Question 1 — The submarine tour

(8 marks)

On a tour of a nuclear submarine, the guide points out four parts of the reactor: the *moderator*, the *control rods*, the *coolant loop* and the *shielding*.

- (a) State the job of each of the four components, one line each. [4]
- (b) The moderator and the control rods both “deal with neutrons” and are the classic mix-up. State the crucial difference between what each does to a neutron. [2]
- (c) Explain why the fuel is *enriched* before it ever enters the reactor, and which isotope the enrichment increases. [2]

Question 2 — One neutron’s family tree

(5 marks)

Each fission of a uranium-235 nucleus releases two or three fast neutrons.

- (a) Explain how this allows a chain reaction, and what *critical mass* means for sustaining it. [3]
- (b) Explain why the released neutrons must be *slowed* before they can reliably cause further fissions in U-235. [2]

Question 3 — Weighing the missing mass

(7 marks)

In one typical fission event, the products’ total mass is about 0.20 u less than the reactants’ total mass.

- (a) State what has become of the missing mass. [1]
- (b) Calculate the energy released by this single event, in joules. [3]
- (c) A student compares fuels by dividing energy released by the mass *converted*, and finds fission and fusion identical (“both give c^2 per kilogram, of course”). State the fair comparison to make instead, and which process wins it. [3]

Question 4 — The town-hall meeting

(4 marks)

Your town debates a small nuclear power station.

- (a) Give one physics-based advantage of fission power over fossil fuels, with a supporting reason. [2]
- (b) Give one genuine physics-based challenge of fission power, with a supporting reason. [2]

Answers

Question 1 — The submarine tour

- (a) *Moderator*: slows the fast neutrons down to thermal speeds. *Control rods*: absorb neutrons, setting or halting the reaction rate. *Coolant loop*: carries thermal energy away from the core to the heat exchanger and turbines. *Shielding*: absorbs the emitted radiation, protecting the crew.
- (b) The moderator *slows* neutrons but keeps them in play, so they can still cause fission. The control rods *remove* them altogether by absorption. Slow down against take out: that is the whole distinction.
- (c) Natural uranium is overwhelmingly U-238, which does not fission readily with thermal neutrons. Enrichment raises the proportion of *uranium-235*, the isotope that does, so that enough fissions occur to sustain the reaction.

Question 2 — One neutron's family tree

- (a) Each fission releases two or three neutrons, and each of those can trigger a further fission, which releases more again. If on average at least one neutron per fission goes on to cause another, the process sustains itself rather than dying out. *Critical mass* is the minimum mass of fuel for which that condition holds: below it, too many neutrons escape through the surface before meeting another nucleus, because the surface-to-volume ratio is too large.
- (b) Fast neutrons are likely to scatter off nuclei or be captured by U-238 without causing fission. The fission cross-section of U-235, meaning the effective target area it presents, is far larger for slow thermal neutrons, so slowing them dramatically raises the chance that a neutron produces another fission.

Question 3 — Weighing the missing mass

- (a) It has been converted into energy, appearing as kinetic energy of the fragments and neutrons and ultimately as thermal energy, in accordance with $E = mc^2$.
- (b) $\Delta m = 0.20 \times 1.66 \times 10^{-27} = 3.32 \times 10^{-28}$ kg, so $E = \Delta mc^2 = (3.32 \times 10^{-28})(3.0 \times 10^8)^2 = 3.0 \times 10^{-11}$ J, about 190 MeV.
- (c) Dividing by the mass *converted* is circular: by $E = \Delta mc^2$ that ratio is c^2 for any process whatever, so it cannot distinguish them. The fair comparison is the energy released per unit mass of *fuel*, that is, of the reactants you actually have to supply. Fusion converts a larger *fraction* of its fuel mass than fission does, so fusion wins per kilogram of fuel.

Question 4 — The town-hall meeting

- (a) For example, the energy density is enormous: a kilogram of nuclear fuel yields of order a million times the energy of a kilogram of fossil fuel, because nuclear binding energies dwarf chemical bond energies. Hence a very small fuel mass, and no carbon dioxide released at the point of generation.
- (b) For example, the fission products are radioactive with half-lives ranging from years to many thousands of years, so the waste must be isolated for far longer than any institution has yet existed. (Other acceptable answers: the cost and hazard of decommissioning; the fact that although accidents are rare their consequences are concentrated and long-lived.)